Nexus between Inflation and Fiscal Deficit in India: Auto Regressive Distributed Lag Model Analysis

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ABSTRACT

The fiscal deficit and inflation have become significant concerns for economists and policymakers, prompting numerous studies on these twin issues in developed and developing economies. However, the results of these studies have been inconclusive due to variations in estimation techniques, periods, and selected variables. To address this, the researcher examined the fiscal deficit-inflation relationship in India from 1990-91 to 2021-22 using the Autoregressive Distributed Lag (ARDL) and Nonlinear Autoregressive Distributed Lag (NARDL) approaches. Contrary to expectations, our ARDL findings did not indicate a linear relationship between fiscal deficit and inflation in the Indian context. However, our empirical results using the NARDL model revealed a nonlinear connection between fiscal deficit and inflation in the long run. Additionally, we found no association between money supply and inflation, supporting India's principles of the Fiscal Theory of the Price Level (FTPL). According to the FTPL, public debt, and taxation policies primarily influence the price level, while monetary policy plays an indirect role. Based on these findings, fiscal policymakers in India must focus on reducing fiscal deficits. Simultaneously, the Reserve Bank of India (RBI) should regulate lending interest rates to create a balanced approach using fiscal and monetary policies to control inflation effectively.

Keywords: Fiscal policy, monetary policy, fiscal deficit, Wholesale Price Index, money supply, crude oil prices, Fiscal Theory of the Price Level, interest rate, cointegration, ARDL Approach, NARDL Approach

Introduction

Harmonizing fiscal and monetary policies is crucial for achieving macroeconomic stability in an economy. Failure to do so can adversely impact the overall economy due to the uncertain nature of certain macroeconomic variables (Parida et al., 2001). In both developed and developing economies, the chronic government deficit, also known as the fiscal deficit, and an increase in the price level, often referred to as inflation, have become major concerns (Friedman, 1968; Fischer and Easterly, 1990). A large fiscal deficit increases public debt and exerts inflationary pressure on the economy (Friedman, 1968; Fischer and Easterly, 1990).

According to Friedman (1968), managing the money supply can control the inflation rate in the long term. If monetized, fiscal deficits can only lead to inflation (Sargent and Wallace, 1981). Monetization occurs when the central bank finances the deficit by increasing the money supply, eventually resulting in an inflationary impact in the long run. Fischer and Easterly (1990) highlighted that rapid money supply growth without a fiscal deficit is rare, indicating that the inflation rate is predominantly a fiscal phenomenon. An alternative perspective explored by Miller (1983) suggested that deficits impact inflation dynamics regardless of monetization. Deficit financing can be done through private monetization and crowding out. Non-monetized deficits, financed through bond issuance, can lead to higher interest rates, which crowd out private investment, reduce real output growth, and increase the price level in the economy.

In the Indian context, the fiscal-monetary policy landscape has undergone significant changes since the early 1990s. These changes include the elimination of automatic monetization of the fiscal deficit through the creation of ad hoc treasury bills in 1997 and the prohibition of the Reserve Bank of India (RBI) from purchasing government securities in the primary market under the Fiscal Responsibility and Budget Management (FRBM) Act, 2003 (Raj et al., 2011). While these alterations have transformed the fiscal-monetary policy framework, large fiscal deficits incurred by the central government continue to influence demand management by the RBI, placing the dynamics of the fiscal deficit and inflation at the forefront in India.

Theoretical perspectives on the dynamics of the fiscal deficit and inflation vary. The Keynesian Theory emphasizes the relationship between aggregate demand and supply, suggesting that an aggregate demand relative to aggregate supply increases inflation if the economy is at full employment. The Quantity Theory of Money, or the Monetarist Hypothesis, famously stated by Friedman (1968), suggests that inflation is primarily a monetary phenomenon. In contrast, the Fiscal Theory of the Price Level argues that public debt and taxation policies drive the price level, with monetary policy playing an indirect role (RBI, 2012). Empirical studies examining the relationship between fiscal deficit and inflation have yielded inconclusive results due to variations in estimation techniques, periods, and selection of variables. Some studies support a strong connection between fiscal deficit and inflation (Parida et al., 2021; Hamburger and Zwick, 1981; Sriyana and Ge, 2019), while others deny such a relationship (Dwyer, 1982; Chakraborty and Varma, 2018). In addition to the fiscal deficit, money supply is essential in determining the price level. Studies have shown that money supply influences the inflation rate in India (Rami, 2010; Bhaduri and Durai, 2012). Crude oil prices are also recognized as a major determinant of the domestic price level and output, impacting inflation and output negatively in India (Bhattacharya and Bhattacharyya, 2001; Mohanty and John, 2015). Given the inconclusive results from previous studies, this present study aims to examine the fiscal deficit-inflation nexus in India from 1990-91 to 2021-22 using both the Autoregressive Distributed Lag (ARDL) approach to Cointegration and the Nonlinear Autoregressive Distributed Lag (NARDL) Bounds Testing of Cointegration. The analysis includes macroeconomic variables such as consumer price index (indicating inflation rate), fiscal deficit, fuel oil energy consumption, exchange rate, money supply, and gross domestic product. The study is structured as follows: Section 2 describes the growth pattern of fiscal deficit and inflation in India, Section 3 describes the dataset and variables, Section 4 discusses the research methodology and empirical analysis of both ARDL and NARDL approaches and Section 5 presents the summary and concluding remarks.

2. Fiscal Deficit and Inflation:

Recent Trends The fiscal deficit and inflation phenomena have gained significant importance in formulating government policies in India. Therefore, it is crucial to analyze the trends of these variables within the Indian context. Figure 1 illustrates the movement of the central government's fiscal deficit as a percentage of gross domestic product (GDP) and the consumer price index (CPI) growth rate from 1990-91 to 2021-22.

A high inflation rate of 18.2% was observed during the early phase. The rate gradually decreased in subsequent years but remained in double digits during the early 1990s. The government successfully reduced the inflation rate to a single digit in 1994-95, maintained until the mid-2000s. However, after 2008, an increasing trend was observed, except one year in the middle, leading to almost double-digit inflation in 2010-11.

Regarding the fiscal deficit, it was maintained at a manageable level until the early 1980s. However, from the mid-1980s to the mid-1990s, the deficit became a matter of grave concern for the government. Despite efforts, the deficit nearly reached double digits in 2002-03. To address this issue, the Government of India implemented the Fiscal Responsibility and Budget Management (FRBM) Act in 2003 to regulate the fiscal deficit. Subsequently, the deficit level decreased and remained under control until 2008. However, with the global financial crisis, the deficit once again increased significantly.

Overall, it is evident that there is a positive relationship between fiscal deficit and inflation in most years, except the early 1980s and the mid-2000s. In the early 1980s, factors such as increased fuel prices and shortage of food supply due to deficient agricultural production contributed to inflationary pressure. Although there was a substantial decrease in the fiscal deficit level, inflation remained high during 1991-92 and 1994-95. This trend can be attributed to globalization and the depreciation of the Indian currency against the dollar, which led to capital inflows and inflation. Additionally, a price hike in essential commodities due to inadequate supply further contributed to the rise in inflation levels. Thorough structural reforms played a vital role in controlling inflationary pressures.

Variables and Data Sources This study examines the dynamics of fiscal deficit and inflation in India using annual data from 1990-91 to 2021-22. Other macroeconomic variables, such as fuel prices, money supply, and interest rates, are also analyzed. The consumer price index (CPI) for all commodities indicates inflation, as major policy changes in India have been based on the inflation level. For the interest rate variable, the commercial bank lending rate (LIR) is chosen, as the fiscal deficit negatively influences the lending rate of commercial banks, which, in turn, affects output and inflation. The growth rate of money supply (GrMS) is taken as a measure of money supply. Crude oil prices (OP) are used as a proxy for oil prices. The fiscal deficit of the central government (CFD) is selected as the fiscal deficit variable. Time-series data for these variables are sourced from the Reserve Bank of India (RBI).

To assess the nexus between fiscal deficit and inflation, the study formulates the following log-linear equation:

$$CPI = f(GFD, M3, ER, OIL, GDP, GAP \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Where:

CPI = inflation rate as indicated by consumer price index;

GFD = gross fiscal deficit of the central government;

M3 = money supply M3;

ER = exchange rate as depicted by the movement in the exchange rate index;

OIL = Fuel (Energy) Index; and

GDP_GAP = output gap estimated by applying the HP filter on the annual real GDP. As equation (1) is only in an implicit form, the explicit form of the model could be expressed as the conventional log-log model for the long-run equilibrium inflation function:

$$LNCPI_t = a_0 + a_1 LNGFD + a_2 LNM3 + a_3 LNER + a_4 LNOIL + a_5 LNGDP_{GAP} + u_t \ldots(2)$$

Table I

	LCPI	**LER**	**LGFD**	**LM3**	**LOIL**
Mean	4.33	3.82	1.61	4.15	4.20
Median	4.25	3.82	1.65	4.22	4.20
Maximum	5.32	4.36	2.22	4.47	4.34
Minimum	3.13	2.86	0.92	3.74	3.98

Std. Dev.	0.64	0.35	0.27	0.24	0.10
Skewness	-0.09	-0.66	-0.37	-0.45	-0.44
Kurtosis	1.86	3.31	3.22	1.64	2.07
Jarque-Bera	1.82	2.56	0.80	3.67	2.25
Probability	0.40	0.28	0.67	0.16	0.33

Sources; Authors Calculation

The descriptive statistics of all variables used in the model are provided in Table 1. For examining and determining the statistical behaviour of these variables, descriptive statistics were calculated.

4. Model Specification and Empirical Analysis

4.1. Auto-Regressive Distributed Lag (ARDL) approach to Co-integration.

The ARDL model, introduced and developed by Phillips and Perron (1988) and later revised by Pesaran et al. (2001), examines the relationships among variables in this study. Unlike traditional statistical methods, the ARDL model offers several advantages when assessing cointegration and short/long-run relationships, regardless of whether the variables are integrated of order I(0), I(1), or a combination of both. However, it cannot be applied when the variables are integrated into order two (I(2)).

The inclusion of both I(0) and I(1) variables in the ARDL model is advantageous since time series data tend to be either integrated of order I(1) or I(0). Moreover, the ARDL approach incorporates the short-run impact of variables and their long-run equilibrium through an error correction, facilitating assessing short-run and long-run relationships. Unlike traditional cointegration tests, the ARDL model allows for different lag structures for each variable, enhancing flexibility (Pesaran et al., 2001).

The ARDL method is known for providing consistent results even with small sample sizes, making it less sensitive to sample size than other cointegration techniques (Phillips and Perron, 1988; Pesaran et al., 2001). To apply the ARDL model, it is necessary to test the null hypothesis of no cointegration ($H0: a1 = a2 = a3 = a4 = a5 = a6 = 0$). The F-test is employed for testing the alternative hypothesis, and critical values are obtained from tables provided by Pesaran et al. (2001). The asymptotic distributions of the F-statistics under the null hypothesis are non-standard, regardless of whether the variables are purely I(0), I(1), or mutually cointegrated (Pesaran et al., 2001). Two sets of asymptotic critical values are provided: one assuming all variables are I(0), and the other assuming all variables are I(1). If the calculated F-statistic exceeds the upper bound critical value, the null hypothesis of no cointegration is rejected. Conversely, the null hypothesis is accepted if it falls below the lower bound critical value. When the computed F-statistic lies between the lower and upper bound critical values, no conclusive result can be determined, and the presence of cointegration can be established using the error correction term. The long-run coefficients can be estimated using the ARDL model presented as follows:

$$\boldsymbol{LNCPI_t = a_0 + \sum_{i=0}^{p} a_1 LNCPI_{t-1} + \sum_{i=0}^{p} a_2 LNGFD_{t-1} + \sum_{i=0}^{p} a_3 LNM3_{t-1} + \sum_{i=0}^{p} a_4 LNER_{t-1} + \sum_{i=0}^{p} a_5 LNOIL_{t-1} + \sum_{i=0}^{p} a_6 LNGDP_{t-1} + u_t \ldots\ldots 3}$$

Where, $\boldsymbol{a_0}$ represents intercept; $\boldsymbol{a_1}$ to $\boldsymbol{a_6}$ denote long-run coefficients; p & q are the lags of endogenous and exogenous variables, respectively, and ut stands for the residual term. Further, error correction representation of the ARDL model can be written as:

$$\boldsymbol{\Delta LNCPI_t = b_0 + \sum_{i=0}^{p} b_1 \Delta LNCPI_{t-1} + \sum_{i=0}^{p} b_2 LNGFD_{t-1} + \sum_{i=0}^{p} b_3 \Delta LNM3_{t-1} + \sum_{i=0}^{p} b_4 \Delta LNER_{t-1} + \sum_{i=0}^{p} b_5 \Delta LNOIL_{t-1} + \sum_{i=0}^{p} b_6 \Delta LNGDP_{t-1} + \emptyset ECM_{t-1} + \varepsilon_t \ldots\ldots 4}$$

where, Δ denotes the first difference; b1 to b5 represent short-run coefficients; ECMt - 1 is the error correction term, φ shows the speed of adjustment and et stands for the residual term.

4.2. Empirical Analysis

For determining the unit root properties of the data, we used two traditional tests: the Augmented Dickey-Fuller (ADF) test and Phillips-Perron (PP) test (Phillips, et al., 1988) [The results are provided in Table 2. From Table 2, ADF and PP tests show that LER is stationary at levels, i. e. I (0), and the rest of the variables are stationary at the first difference, i. e. I(I). Therefore, mixed order of integration, i. e. I(0) and I(1), validates the implementation of the ARDL bounds testing approach to cointegration. We used the bounds test for cointegration and the results are reported in Table 3. Since the

Table 2

UNIT ROOT TEST TABLE (PP)

	At Level					
		LCPI	LER	LGFD	LM3	LOIL
With Constant	t-Statistic	-1.6695	-3.0633	-3.0101	-1.1719	-3.2654
	Prob.	0.4366	0.0397	0.0446	0.6742	0.0252
		n0	**	**	n0	**
	At First Difference					
		d(LCPI)	d(LER)	d(LGFD)	d(LM3)	d(LOIL)
With Constant	t-Statistic	-3.1746	-5.2653	-7.1021	-5.091	-5.5633
	Prob.	0.0313	0.0002	0	0.0002	0.0001
		**	***	***	***	***
UNIT ROOT TEST TABLE (ADF)						
	At Level					
		LCPI	LER	LGFD	LM3	LOIL
With Constant	t-Statistic	-0.7605	-3.5472	-3.0101	-1.177	-2.869
	Prob.	0.8142	0.013	0.0446	0.672	0.0602
		n0	**	**	n0	*
At First Difference						
		d(LCPI)	d(LER)	d(LGFD)	d(LM3)	d(LOIL)

With Constant	t-Statistic	-6.6454	- 5.1596	-6.199	-5.0919	-5.5463
	Prob.	0	0.0002	0	0.0002	0.0001
		***	***	***	***	***

Notes: (*) Significant at the 10%; (**)Significant at the 5%; (***) Significant at the 1%. and (no) Not Significant

*MacKinnon (1996) one-sided p-values.

Sources; Authers Calculation

Table 3

F-Bounds Test				
Test Statistic	**Value**	**Signif.**	**I(0)**	**I(1)**
F-statistic	7.81	10%	2.45	3.52
k	4	5%	2.86	4.01
		2.50%	3.25	4.49
		1%	3.74	5.06

Sources; Authers Calculation

Table 4

Residual Diagnostic Test	Statistics	P-Value
Serial Correlation Test	2.24	0,08
Heteroskedosticity Test	0.92	0.51
Normality Test	0.23	78

Sources; Authors Calculation

Table 5 Long Run Estimates of the ARDL Model

Variable	Coefficient	Std. Error	t-Statistic	Prob.

C	**-1.67**	**0.25**	**-6.57**	**0.00**
LER	36.61	317.29	0.12	0.91
LGFD	-8.12	71.79	-0.11	0.91
LM3	9.07	81.32	0.11	0.91
LOIL	-114.64	1053.27	-0.11	0.91
CointEq(1)*	0.005	0.000	6.79	0

Sources; Authors Calculation

Note: ** represent 5 percent level of significance

The obtained F-statistics (7.81) are greater than the upper critical bound value (3.52) at a 5 percent level of significance, leading us to reject the null hypothesis of no cointegration. Consequently, the results indicate that all variables move in the same direction in the long run. To determine the appropriate lag length for the model, the Akaike Information Criterion (AIC) was utilized, and Figure 2 displays that the AIC recommends the ARDL (4, 4, 4, 4, 3) model for evaluating the long-run coefficients. In order to assess the suitability of the model, several diagnostic tests were conducted as shown in Table 4. The Breusch-Godfrey serial correlation LM test and the Breusch-Pagan-Godfrey test demonstrate that the model is free from autocorrelation and heteroskedasticity in the residuals, respectively. The Jarque-Bera statistics indicate that the residual term follows a normal distribution. Moving on to the estimation of the long-run coefficients using the ARDL model, the results are presented in Table 5. It is observed that the coefficients of LER and LM3 are positive, as expected, but they have an insignificant impact on the Consumer Price Index (CPI) at a 5 percent level of significance. This suggests that there is no relationship between LGFD & CPI and LOIL & CPI in the Indian context. On the other hand, the coefficient of LER is negative and has an insignificant impact on CPI at a 5 percent level of significance, implying that a 1 percent increase in LER will result in a 114.64 percentage point decrease in CPI in the long run. Furthermore, LGFD has an insignificant impact on CPI at a 5 percent level of significance. Notably, the coefficient of the error correction term (ECM) is positive and significant at a 5 percent level of significance. A positive ECM indicates that the series will diverge instead of converging, suggesting a lack of long-run equilibrium.

Thus, the results of the ARDL approach found no evidence of linear/symmetric relationship between the fiscal deficit and inflation in the Indian context. Therefore, we decided to apply the Nonlinear Autoregressive Distributed Lag (NARDL) model introduced in order to verify the nonlinear/asymmetric relationship between the variables.

4.3. Nonlinear Autoregressive Distributed Lag (NARDL) Bounds Testing of Co-integration

The study has considered the general form of the NARDL model to test the asymmetric co-integration:

$$CPI_t = \alpha_0 + \beta_1^+ GFD_t^+ + \beta_2^- GFD_t^- + \beta_3^+ M3_t^+ + \beta_4^+ M3_t^- + \beta_5^+ ER_t^+ + \beta_6^+ ER_t^- + \beta_7^+ OIL_t^+ + \beta_8^+ OIL_t^- + \varepsilon_t$$

…………………………………1

In equation (4), GFD_t^+ & GFD_t^-; $M3_t^+$ & $M3_t^-$; ER_t^+ & ER_t^- and $\beta_7^+ OIL_t^+$ & $\beta_8^+ OIL_t^-$ are the partial sum of positive and negative changes in fiscal deficit, money supply, exchange rate, and lending fuel oil energy, respectively:

$GFD_t^+ = \sum_{j=1}^{t} GFD_t^+ = \sum_{j=1}^{t} Max\Delta GFD_j \ ,0$ ……………………..5

$GFD_t^- = \sum_{j=1}^{t} GFD_t^- = \sum_{j=1}^{t} Min\Delta GFD_j \ ,0$ ……………………..6

$M3_t^+ = \sum_{j=1}^{t} M3_t^+ = \sum_{j=1}^{t} Max\Delta M3_j \ ,0$ ……………………..7

$M3_t^- = \sum_{j=1}^{t} M3_t^- = \sum_{j=1}^{t} Min\Delta M3_j \ ,0$ ……………………..8

$ER_t^+ = \sum_{j=1}^{t} ER_t^+ = \sum_{j=1}^{t} Max\Delta ER_j \ ,0$ ……………………..9

$ER_t^- = \sum_{j=1}^{t} ER_t^- = \sum_{j=1}^{t} Min\Delta ER_j \ ,0$ ……………………..10

$OIL_t^+ = \sum_{j=1}^{t} OIL_t^+ = \sum_{j=1}^{t} Max\Delta OIL_j \ ,0$ ……………………..11

$OIL_t^- = \sum_{j=1}^{t} OIL_t^- = \sum_{j=1}^{t} Min\Delta OIL_j \ ,0$ ……………………..12

has modified equation (4); thus, we re-arranged the ARDL form along the line of and as:

$$\Delta CPI_t = \rho CPI_{t-1} + \delta_1^+ GFD_t^+ + \delta_2^- GFD_t^- + \delta_3^+ M3_t^+ + \delta_4^+ M3_t^- + \delta_5^+ ER_t^+ + \delta_6^+ ER_t^- + \delta_7^+ OIL_t^+ + \delta_8^+ OIL_t^- + \sum_{j=1}^{p-1} \emptyset_j \, CPI_{t-1} \sum_{j=0}^{q-1} [\vartheta_1^+ GFD_t^+ + \vartheta_2^- GFD_t^- + \pi_3^+ M3_t^+ + \pi_4^+ M3_t^- + \sigma_5^+ ER_t^+ + \sigma_6^+ ER_t^- + \tau_8^+ OIL_t^+ + \tau_8^- OIL_t^-] + \varepsilon_t$$

…………….13

In equation (13), *p* and *q* are the lag order of endogenous and exogenous variables, respectively. The coefficients (ω+1 , ω-2 , ω+3 , ω-4 , ω+5 , ω-6 , ω+7 , ω-8) represent the long-run relationship, whereas coefficients. represent the short-term dynamics of the model. However

$\beta_1^+ = (-\frac{\omega_1^+}{p})$, $\beta_2^+ = (-\frac{\omega_2^+}{p})$, $\beta_3^+ = (-\frac{\omega_3^+}{p})$, $\beta_4^+ = (-\frac{\omega_4^+}{p})$, $\beta_4^+ = (-\frac{\omega_4^+}{p})$, $\beta_5^+ = (-\frac{\omega_5^+}{p})$,

$\beta_6^+ = (-\frac{\omega_6^+}{p})$, $\beta_7^+ = (-\frac{\omega_7^+}{p})$, $\beta_8^+ = (-\frac{\omega_8^+}{p})$…………………………………14

are the asymmetric long-run elasticities for *GFD*+ ,G*FD*- , *M3*+ , *M3*- , *ER* + , *ER* - , *OIL*+ & *OIL*- , respectively, on CAD. e*t* is the white noise (i. e., error term normally distributed around zero mean and constant variance). For estimating asymmetric co-integration among the variables, the joint null hypothesis of no co-integration, r = ω+1 = ω-2 = ω+3 = ω-4 = ω+ 5 = = ω-6 = ω+7 = ω-8 = 0 is tested using the bounds testing approach based on the F-statistics. If the value of the F -statistics is larger than the upper bound critical value, it confirms the existence of long-run relationship between the variables; if the value is below the lower bound, it negates any relationship. The test is inconclusive if the value of the *F*-statistics lies between the two bounds. To estimate the short- and long-run asymmetric effect, we adopted the Wald statistics.

4.4. Empirical Analysis

For the implementation of the NARDL approach, it is mandatory to verify the order of integration of all variables. From Table 2, *ADF* and *PP* tests show that *M3* is stationary at levels, i. e. I(0), and the rest of the variables are stationary at first difference, i. e. I(1). Therefore, different order of integration, i. e. I(0) and I(1), validates the implementation of the NARDL approach to co-integration. The study has used the FPSS statistic of to detect asymmetric co-integration among the variables; the results are reported in Table 6. This test provides reliable results on asymmetric co-integration by including different orders of assimilation. Since the calculated value of the FPSS statistics (4.32) is greater than the upper bounds critical values at 5 percent level of significance, we rejected the null hypothesis of no asymmetric cointegration among the variables. Before estimating the NARDL model, we con ducted some diagnostic tests to check the suitability of the model (see table 7). The Breusch-Godfrey serial correlation LM test and the Breusch-PaganGodfrey test reveal that the model is free from the problem of autocorrelation and heteroskedasticity in the residuals, respectively. The Jarque-Bera statistics show that the residual term is normally distributed. Further, we employed the Wald test to scrutnise both long-run and short-run asymmetries to validate the NARDL approach and the outcomes of the long-and short-run asymmetries are reported in Table 8. The results support the existence of long-run asymmetries between the negative and positive parts of fiscal deficit (*GFD*) and oil consumption (*OIL*). Furthermore, the findings of short run asymmetry reveal that the Wald test failed to reject the null hypothesis of short-run symmetries between the negative and positive parts of all variables. Following the general to specific method, the study estimates the NARDL model shown in equation (13); the results of long-run estimates are reported in Table 9. The results show that the influence of positive and negative changes in *GFD* on *CPI* appears to be positive, and it is significant at 5 and 10 percent level of significance, respectively. This result is similar to other studies, including. The obtained results indicate that a 1 percent increase in *FD* will lead to a 4.70 percentage point increase in *CPI*, whereas a 1 percent decrease in *FD* will lead to a 3.93 percentage point decrease in *CPI*. However, the positive change in *FD* has a larger influence on *CPI* compared to the negative change in *FD*. Another essential element in determining the price level is the growth rate of the money supply. As shown in Table 9, positive and negative parts of M3 have no asymmetries in the long run as well as in the short run. The impact of positive and negative change in *M3* on *WPI* is insignificant, indicating no direct relationship. This result is contradictory to the theory of monetarists, which argued that inflation is always a monetary phenomenon. The above-mentioned findings, i. e., a positive and significant relationship between fiscal deficit and inflation and no association between money supply and inflation support the fiscal theory of the price level in India. According to this theoretical paradigm, the fiscal deficit is the main factor responsible for influencing inflation, and money supply plays no significant role in determining inflation. The dominance of the fiscal policy over monetary policy along with an increase in government liabilities restrict the commitment of the Central Bank to achieve price-level stability. The next important factor considered in the model for determining inflation is oil prices. We found that positive changes in *OP* have a positive and significant impact on inflation at 5 percent level of significance, whereas a negative change in *OP* has a negative but insignificant impact on inflation at 5 percent level of significance. This indicates that a 1 percent increase in *OP* will lead to a 0.49 percentage point increase in *CPPI* in the long run. The result obtained here is in line with results presented by etc. Interest rate also plays a vital role in determining the price level in the economy. The positive and negative parts of interest rate failed to reject the null hypothesis of no asymmetries both in the long-run and short-run. We found that a positive change in *ER* appears to hurt CPI. Still, it is insignificant at a 5 percent level of significance, whereas a negative change in *ER* hurts *CPI* that is significant at a 10 percent level of significance. This implies that a 1 percent decrease in *ER* will lead to a 1.23 percentage point increase in *WPI* in the long run. This supports Keynes' version of the quantity theory of money, which argued that the central bank makes policy changes by increasing interest rates to control the economy's price level. Lastly, the verification of the short-term dynamics is also important. The error correction term (*ECM*) is negatively significant at a 5 percent significance level, confirming the long-run relationship between *WPI* and its covariates. Here, the coefficient is -0.62, which shows that the speed of adjustment toward long-run equilibrium is 62 percent annually. Concisely, the study concludes that the assumption of a symmetrical relation between fiscal deficit and inflation could lead to misleading inferences. Notable, both negative and positive changes in the fiscal deficit have a differential impact on inflation in the long run.

Table 6

Equation	F_{PSS}	Level of significance	Critical Value		Result
			Lower Bounds	Upper Bound	
CPI=f(*GFD*+ ,*GFD*- , *M3*+ , *M3*- , *ER* + , *ER* - , *OIL*+ & *OIL*-)	4.32	5 percent	2.75	3.98	Co-integration
		10 percent	2.23	4.12	Co-integration

Table 7 Diagnostic Tests

Residual Diagnostic Test	Statistics	P-Value
serial Correlation test	26.15	0.10
Heteroskedasticity test	0.64	0.4
Normality Test	0.49	0.83

Table 8 Asymmetry Statistics

Exogenous Variable	Long run Asymmetry		Short run Asymmetry	
	F- test	P-value	F-test	P-value
GFD	12.56**	0.00	0.96	0.39
ER	4.36	0.09	3.08	0.15
OIL	1.09	0.39	0.37	0.59
M3	2.70	0.14	0.07	0.86

Table 9 Long-Run Coefficients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(GFD)	0.			
D(GFD(-)				
D(LER)	0.05	0.07	0.72	0.02
D(LER(-1))	0.10	0.07	1.41	0.00
D(LM3)	-0.04	0.09	-0.46	0.04
D(LM3(-1))	0.03	0.09	0.36	0.56
D(LOIL)	0.71	0.40	1.79	0.98

D(LOIL(-1))	0.33	0.39	0.87	0.40
ECM(-1)*	-0.04	0.01	-3.14	0.01

5. Conclusion and Suggestions

The present study estimates the impact of the fiscal deficit on inflation in India using annual datasets from 1980–81 to 2016–17. The study has considered additional macroeconomic variables, namely, money supply, oil prices, and lending interest rates, as the determinants of inflation. For empirical analysis, we used the ARDL and NARDL approaches. To check the stationarity of all macroeconomic variables, we applied the Augmented Dickey-Fuller (*ADF*) test and Phillips-Perron (*PP*) test. ADF and PP tests showed that GrMS is stationary at levels, i.e... I (0), and the rest of the variables are fixed at the first difference, i. e. I(I) allowed us to apply the ARDL and NARDL models. The ARDL approach found no evidence of a linear relationship between fiscal deficit and inflation in the Indian context. Therefore, we used the NARDL model to verify the nonlinear/asymmetric relationship between the variables. The NARDL approach distinguishes the positive and negative changes in explanatory variables for estimating the response in the dependent variable. We obtained the following results after using the NARDL model. First, the bounds test confirmed the asymmetric cointegration among the selected variables since the FPSS statistic is higher than the critical upper bound value at a 5 percent significance level. Second, the Wald test found asymmetries between the negative and positive parts of the fiscal deficit (*GFD*) and oil prices (*OP*) in the long run only. Third, the impact of positive and negative changes in *FD* on *CPI* appears to be positive and significant. These results are akin to those of Parida et al., (2001). The impacts of positive and negative parts of the money supply have no asymmetries in both the long run and short run. The effect of positive and negative changes in money supply on CPI is insignificant, indicating no direct relationship. This result contradicts the theory of monetarists, which argues that inflation always and everywhere a monetary phenomenon. The findings mentioned above, i. e., a positive and significant relationship between fiscal deficit and inflation and no association between money supply and inflation support the fiscal theory of the price level in India's case. Fourth, the impact of both positive and negative changes in oil prices on WPI is positive and significant. This result supports many theoretical and empirical studies, including pearson, et. al., (2001) Lastly, the coefficient of *ECM* is 0.04, meaning that the speed of adjustment towards long-run equilibrium is 4 percent annually. Fiscal policy plays a vital role in controlling price fluctuations by contractionary and expansionary fiscal policy. Presently, India is facing recession due to depreciation of the rupee, high interest payment on borrowings and increase in oil and petrol prices in the international market, resulting in a decline in industrial production and, hence, increase in unemployment. Therefore, this study suggests that the government should adopt expansionary fiscal policy by reducing taxes for the manufacturing sector and investing in infrastructure, electricity, health, education, to boost the demand, which will stimulate private players to produce more. This action will have a long-term stabilising effect on the Indian economy. It will increase production results and purchasing power by providing people with employment, decrease inflationary pressure by reducing the demand and supply mismatch, and help curb inflation.

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